

Sai Praneeth Kathi Moksha Gnana

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PROFESSIONAL SUMMARY

Data Engineer and MS Information Systems candidate at UMD Smith building cloud data pipelines, ingestion workflows, validation frameworks, and analytics platforms with Python, SQL, BigQuery, Snowflake, Airflow, Kafka, Docker, FastAPI, and GCP/Azure. Delivered retail, forensic, and healthcare solutions spanning data integration, orchestration, schema design, quality checks, automation, and cloud deployment.

SKILLS

Programming & Databases: Python, SQL, JavaScript, FastAPI, BigQuery, Snowflake, MySQL, MongoDB, SQLite

Data Engineering & Orchestration: Airflow, Kafka, dbt, Docker, PySpark, ETL/ELT, Data Validation

Cloud & Automation: GCP, Azure, Cloud Run, Copilot Studio, Power Automate, Database

Analytics & ML Libraries: Pandas, NumPy, Scikit-learn, XGBoost, Random Forest, Streamlit

AI & Dev Tools: LangChain, LlamaIndex, ChromaDB, FAISS, RAG, LLM APIs, Git, GitHub, JIRA

EDUCATION

University of Maryland, Robert H. Smith School of Business

Master of Science, Information Systems and AI,

Dec 2026

College Park, MD

Dayananda Sagar University

Bachelor of Technology, Electronics and Communications Engineering

Oct 2023

Bengaluru, India

EXPERIENCE

University of Maryland, Robert H. Smith School of Business

Sep 2026 - Present

Research Assistant, Smith Behavioral Lab

College Park, MD, United States

- Facilitate behavioral studies and survey-based experiments for faculty researchers in the Smith Behavioral lab, supporting structured data collection from undergraduate participants across multiple research projects.
- Monitor participant workflows and study sessions to ensure protocol adherence, data completeness, and consistent execution for reliable downstream analysis.
- Coordinate participant logistics and research documentation while identifying and escalating data collection or process issues affecting study quality.

University of Maryland, Robert H. Smith School of Business

Jun 2026 - Present

Graduate Teaching Assistant, BMSO 758L - AI and Smart Agents

College Park, MD, United States

- Provide technical support for Python-based NLP and agentic AI assignments involving text preprocessing, sentence-transformer embeddings, cosine similarity, OpenAI/Anthropic APIs, prompt engineering, RAG, and multi-agent workflows.
- Review and troubleshoot Jupyter/Google Colab projects using pandas, NumPy, NLTK, spaCy, scikit-learn, Streamlit, ChromaDB/FAISS, and LangChain/LlamaIndex, validating code quality, reproducibility, and analytical outputs.

Microsoft | UMD Smith Capstone

Jan 2026 - Present

AI & Data Analytics Consultant | Microsoft-Sponsored Capstone

College Park, MD, United States

- Designed a five-agent EMS coordination system spanning four workflow stages and 13 core data entities, producing the ERD, DFD, CRUD matrix, data-warehouse requirements, and Azure/Copilot Studio migration architecture.
- Built and deployed a React/Vite, FastAPI, SQLite, Ollama, and Whisper prototype supporting live transcription, structured extraction, facility routing, human approvals, SBAR generation, audit trails, and FHIR/PDF exports.

Mathco

Jan 2024 - Jul 2025

Data Analyst

Bengaluru, India

- Standardized extracts from 20+ data vendors and competitor retail channels, including Amazon, Costco, Best Buy, and Wayfair, using Python for data wrangling and automation by resolving naming inconsistencies and converting 4+ date formats into a unified BigQuery schema.
- Engineered a SQL and Python data validation layer with row-count checks, client-defined business rules, and QA logging across competitor treatments, reducing pipeline errors by 60% and improving data completeness for analysis.
- Automated nightly SQL workflows to refresh vendor data and generate time-series forecasts in Google Cloud Platform, reducing reporting turnaround from 2 days to 6 hours.
- Queried and reconciled Fortune 1 retail client data from 20+ third-party market data vendors using advanced SQL window functions and period-over-period analysis, identifying \$20M in stakeholder-validated growth opportunities across 6+ business segments.
- Developed Tableau executive dashboards delivering YTD, market share, and performance metrics benchmarked against historical baselines for the client's Customer Insights and Strategy team.

PROJECTS

RetailIQ: Retail Analytics Platform | <https://retailiq-demand-intelligence-420746557396.us-central1.run.app/>

- Architected a Snowflake-backed platform processing 421K sales rows across 45 stores; automated a 9-step Airflow DAG, containerized Kafka streaming with Docker, and deployed the Streamlit application on Google Cloud Run.

Forensic Analytics Dashboard | <https://eyforensic.streamlit.app/>

- Developed a Streamlit forensic analytics dashboard for EY's Swindle in the Age of AI challenge, identifying timing violations in 45.2% of invoices and pre-release anomalies in 818 of 4,673 line items, earning Best Data Scientist.

NexGen Agentic Platform | <https://nexgen-ai.streamlit.app/>

- Led technical development for a four-member team, building a full-stack agentic AI platform using Streamlit, LangChain, LLM agents, and a RAG pipeline with vector-based retrieval; earned Honorable Mention at the 2025 Agentic AI Case Competition.

AWARDS & HONORS

- Best Data Scientist Award (2026):** EY x UMD Swindle in the Age of AI Challenge
- Honorable Mention (2025):** Agentic AI Case Competition, NexGen Agentic Platform
- Project Impact Recognition Kudos (2025):** Mathco, for ownership, QA rigor, and automation of Walmart retail workflows